ASCOMP 2026 POSTER

Min-Marginal Power Indices for Multi-Objective Feature Selection

SCAN ME

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01 CORE IDEA & ALGORITHM

From average importance to worst-case evidence.
A harmful coalition witness: adding a feature turns v = 1 into 0.

$$v_\alpha(S) = 1[U_{val}(S) \geq \alpha\, U_{val}(N)]$$

$$\mathbf{MinPI_i = \min [v_\alpha(S \cup \{i\}) - v_\alpha(S)]}$$

over $S \subseteq N \setminus \{i\}$; N = analyzed features

ALGORITHM FLOW

1 Prepare the game

Split 64% / 16% / 20% into train / validation / test.
Fit coalition models on train; evaluate the game on validation.

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2 Estimate MinPI & collect witnesses

A: sample coalition size, then a subset, uniformly.
B: perturb a training-only MI-correlation seed
(drop each member with p = 0.3; add 0-5 features).
Take the minimum observed marginal; H = features with estimate -1.

n ≤ 20: exact | 20 < n ≤ 75: B* | n > 75: min(A, B)
*Use A if correlation neighbors are unavailable; n = |N|.

↓

3 Validate joint removal

$$\mathbf{U_{val}(N \setminus H) \geq U_{val}(N) - \varepsilon}$$

ε = 0.005 (classification); 0.01 (regression).
PASS: remove all H. **FAIL:** keep all features.

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4 Evaluate (optional NSGA-II search)

Select on validation; optionally run MinPI-prefiltered NSGA-II.
Finally report held-out utility and feature compression.

02 RESULTS & EVIDENCE

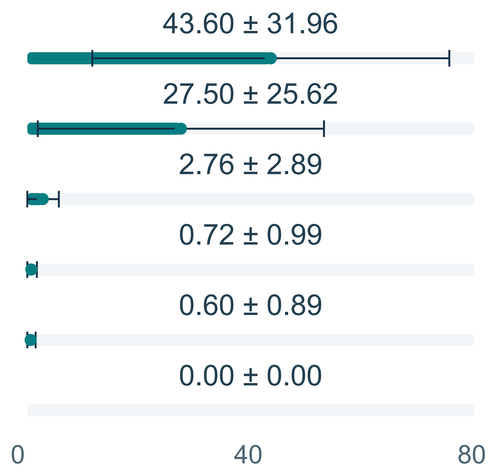
6 datasets × 5 seeds | XGBoost

α = 0.95; training-only preprocessing; ≤100 analyzed features.
Utilities: ROC AUC, OVR AUC or R^2.

12.53%	-0.000313
mean feature removal	mean held-out ΔU

Five-seed mean ± SD; removal of analyzed features.
ΔU = held-out utility of retained subset - full-model utility.

DATASET	REMOVED (%)	HELD-OUT ΔU
Madelon	43.60 ± 31.96	+0.00469 ± 0.00497
Abalone	27.50 ± 25.62	-0.00565 ± 0.01077
Credit-Card-Fraud	2.76 ± 2.89	-0.00078 ± 0.00335
Spambase	0.72 ± 0.99	-0.00003 ± 0.00007
Gisette	0.60 ± 0.89	-0.00011 ± 0.00016
ISOLET	0.00 ± 0.00	0.00000 ± 0.00000



MATCHED-COUNT REMOVAL: MinPI vs random (ΔU)
Madelon: +0.0047 vs -0.0732
Abalone: -0.0056 vs -0.0370

NSGA-II: feature count vs predictive utility

800 evaluations/variant; mean held-out ΔHV (hypervolume) +0.01584.
Madelon: +0.09515; overall Holm p = 0.4375.